

Smart Visual Crop Advisory System

AI-Powered Foliar Disease Diagnosis, Agro-Meteorological Risk Analysis, and Multilingual Voice Assistance (Kannada, Hindi & English)

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Institution:	Department of CSE, Sir M. Visvesvaraya Institute of Technology, Bengaluru, Karnataka, India
Core Model:	MobileNetV2 Transfer Learning (97.07% Val Accuracy) + Grad-CAM XAI
Dataset:	PlantVillage Dataset (20,639+ Images across 15 Crop Disease Classes)
Languages:	Kannada (kn-IN), Hindi (hi-IN), English (en-IN)
Tech Stack:	FastAPI, PyTorch, Torchvision, Open-Meteo, React 18, Vite, TailwindCSS, Web Speech API

Executive Summary:

Smallholder farmers in South Asia face severe crop losses due to delayed diagnosis, illiteracy, and language barriers. This project presents an end-to-end intelligent agricultural decision-support web application that combines deep learning visual classification (MobileNetV2 with Grad-CAM heatmaps), real-time microclimatic disease progression modeling (Open-Meteo weather intelligence), a 4-pillar structured agronomic knowledge base (Root Cause, Organic Treatment, Chemical Dosages with PHI, and Cultural Prevention), and an interactive voice assistant offering 100% native voice synthesis and speech recognition in Kannada, Hindi, and English.

1. System Architecture & Modular Design

Module	Core Responsibility	Key Technologies
Module A: Vision & XAI	Leaf image ingestion, MobileNetV2 inference (97.07% accuracy), and Grad-CAM saliency heatmaps.	PyTorch, Torchvision, OpenCV, PIL
Module B: Weather Risk	Live microclimate ingestion via GPS/Open-Meteo. Continuous 0–100 Environmental Risk Scoring.	Open-Meteo API, Python Agro Engine
Module C: 4-Pillar Advisory	Root Cause, Organic Remedies, Exact Chemical Formulations & Dosages with PHI, and Prevention.	Multilingual JSON Knowledge Base (kn, hi, en)
Module D: Voice Engine	100% native TTS audio playback and natural STT voice query parsing in Kannada, Hindi, and English.	Web Speech API (TTS/STT), Intent Parser

2. Deep Learning Vision Engine & PlantVillage Training

The computer vision backbone uses **MobileNetV2** transfer learning with inverted residual blocks. Trained locally on an **NVIDIA GeForce RTX 4060 Laptop GPU** across 20,639 PlantVillage leaf images, the model achieved **97.07% validation accuracy** within 4 epochs with sub-35ms GPU latency.

- **Class-Weighted CrossEntropyLoss:** Mathematically penalizes errors inversely proportional to class frequency, solving severe class imbalance.
- **Explainable AI (Grad-CAM):** Generates gradient-weighted saliency heatmaps confirming focus on actual disease lesions.
- **Multi-Leaf Generalization:** Validated across distinct crops (Tomato, Potato, Pepper) producing high-confidence, differentiated diagnoses.

3. Agro-Meteorological Environmental Risk Engine

Continuous Risk Scoring Formula:

$$Risk\ Score = Base\ Severity + f(Humidity) + f(Temperature) + f(Rain) + f(Wind)$$

- **Base Severity:** Critical = 40 pts, Moderate = 20 pts, Healthy = 0 pts.
- **Humidity:** +25 pts if $RH \geq \text{Pathogen Min Threshold}$; +12 pts if within 10% deficit.
- **Temperature:** +25 pts if $Temp \in [T_{min}, T_{max}]$; +12 pts if near optimal ($\pm 3^{\circ}C$).
- **Rain & Wind:** Rain > 0 mm (+10 pts leaf wetness); Wind > 12 km/h (+8 pts spore dispersal).

4. Operational Run & Deployment Commands

1. Launch Backend Server:

```
cd backend && venv\Scripts\activate && python run.py
```

2. Launch Frontend Application:

```
cd frontend && npm run dev (Open http://localhost:5173)
```

3. Train MobileNetV2 on PlantVillage Dataset (RTX 4060 GPU):

```
python ml\train.py --data_dir "D:\major project\PlantVillage\PlantVillage\PlantVillage" --epochs 15 --batch_size 32
```

5. Project Team & Acknowledgments

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